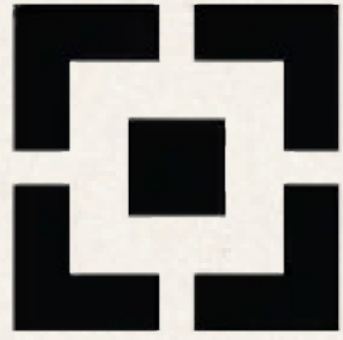
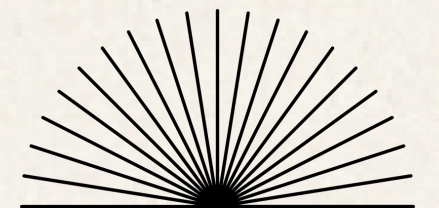




HDFC AI based Load Prediction System

AI-driven classification to automate loan approvals
and enhance risk assessment accuracy

Powered By: AI



Problem Statement

To develop a supervised machine learning model that predicts loan approval status based on applicant financial and personal details, helping improve loan processing speed and decision accuracy.

According to recent reports of HDFC Bank, the Gross NPA ratio is approximately 1.2%–1.4% and the Net NPA ratio is around 0.4%–0.5%, indicating measurable credit risk. Hence, a supervised machine learning model is required to improve loan approval decisions and reduce future defaults.

Challenges

- Time-Consuming Processing
- Class imbalanced problem
- Feature Complexity

Solution at a Glance

- **Automated Loan Pre-Screening using Machine Learning**

Applicant data is preprocessed and transformed through feature engineering to enable intelligent credit evaluation.

- **Supervised Classification for Credit Risk Prediction**

Models such as Logistic Regression, Random Forest, and XGBoost are trained and validated using performance metrics (Accuracy, Precision, Recall, F1-Score).

- **Data-Driven NPA Risk Minimization**

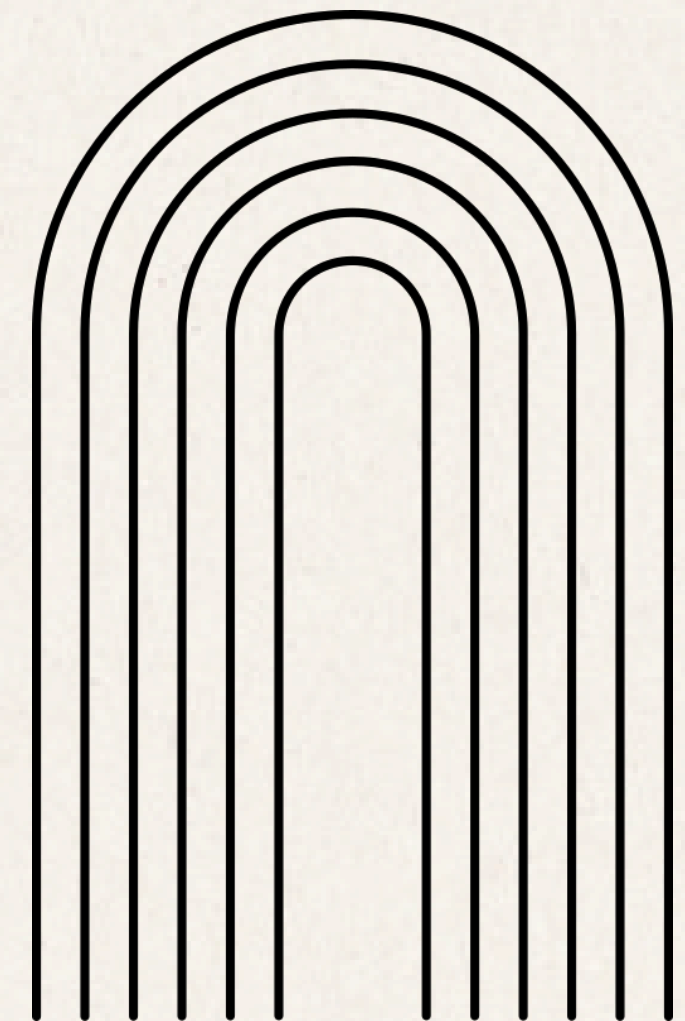
Predictive modeling helps identify high-risk applicants in advance, supporting reduction of default probability and improving asset quality.

Objectives

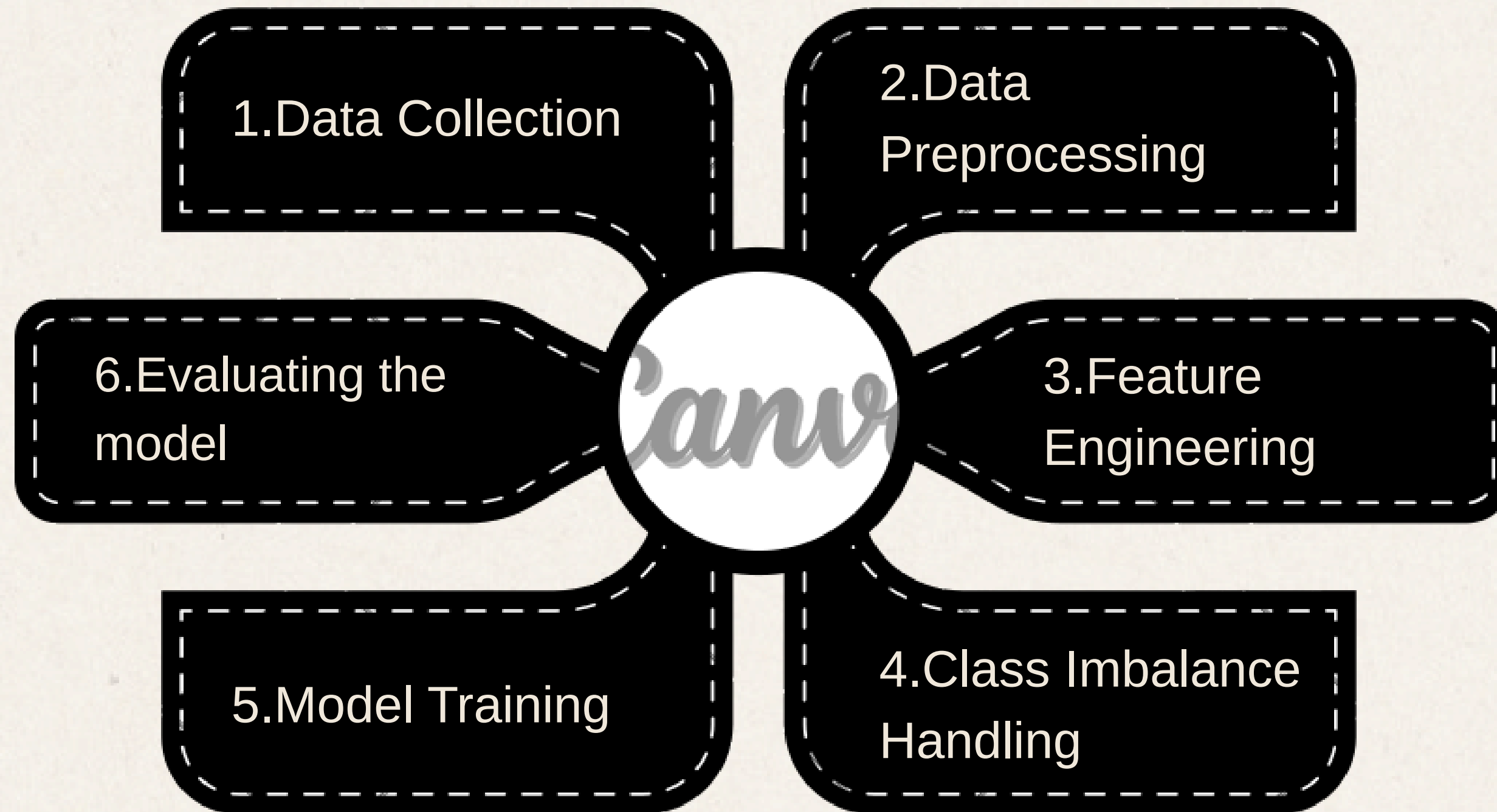
- Build a supervised machine learning model to classify loan applications as Approved or Rejected.
- Improve accuracy using proper data preprocessing and choosing right algo.
- Automate loan screening to reduce manual effort and processing time.

Goals

- Faster loan approval turnaround time.
- Reduce default risk through better risk prediction.
- Improve customer experience and operational efficiency.



Working of the Model



DATASET DESCRIPTION

Source

The dataset consists of historical loan application records collected from publicly available financial datasets and banking-related data repositories (such as Kaggle and open ML datasets) for academic and research purposes.

Dataset Overview

- **Total Records:** 45k loan applications
- **Total Features:** 13 attributes
- **Target Variable:** Loan_Status (Approved / Rejected)
- **Problem Type:** Binary Classification

Class Distribution

Approved: ~68%

Rejected: ~32%

This indicates a moderate class imbalance, where approved loans significantly outnumber rejected ones. Such imbalance may bias the model toward the majority class if not handled properly.

Feature Categories

- Demographic Attributes – Gender, Marital Status, Dependents, Education
- Financial Attributes – Applicant Income, Co-applicant Income, Loan Amount, Loan Term
- Credit Risk Indicators – Credit History, Employment Status, Property Area

By analyzing these features, the proposed supervised machine learning model learns patterns associated with loan approval decisions, enabling data-driven credit risk assessment and potential reduction in future NPAs.

Data Preparation

1.Data Collection

- Historical loan application records.
- Applicant financial and demographic details.
- Target variable: Loan_Status (Approved/Rejected).
- Structured dataset used for model training.

2.Data Cleaning

Missing Values:

- Categorical: Mode Imputation
- Numerical: Median Imputation
- Outliers: Capping extreme values in LoanAmount

3.Feature Engineering

- Standard Feature engineering present.
- Missing values are handled.
- Encoding categorical variables.
- Advanced feature engineering is not done.

4.Imbalance Handling

- Standard imbalanced handling is done
- Stratified Sampling is used, equal yes, no.
- Advanced imbalanced handling is not used like SMOTE, Overhandling, Undersampling.

5. Model Training

- Model trained accordingly.
- Trained multiple classifiers
- Compared performance metrics
- Selected best model

Machine Learning models used

- Logistic regression
- Random Forest
- XG Boost

Why Other Models Were Not Selected?

- Dataset Size Limitation – Complex models like Neural Networks require very large datasets for better performance.
- Overfitting Risk – Highly complex models may overfit on structured/tabular data.
- Interpretability Issues – Simpler models like Logistic Regression provide better explainability compared to black-box models.

LOGISTIC REGRESSION

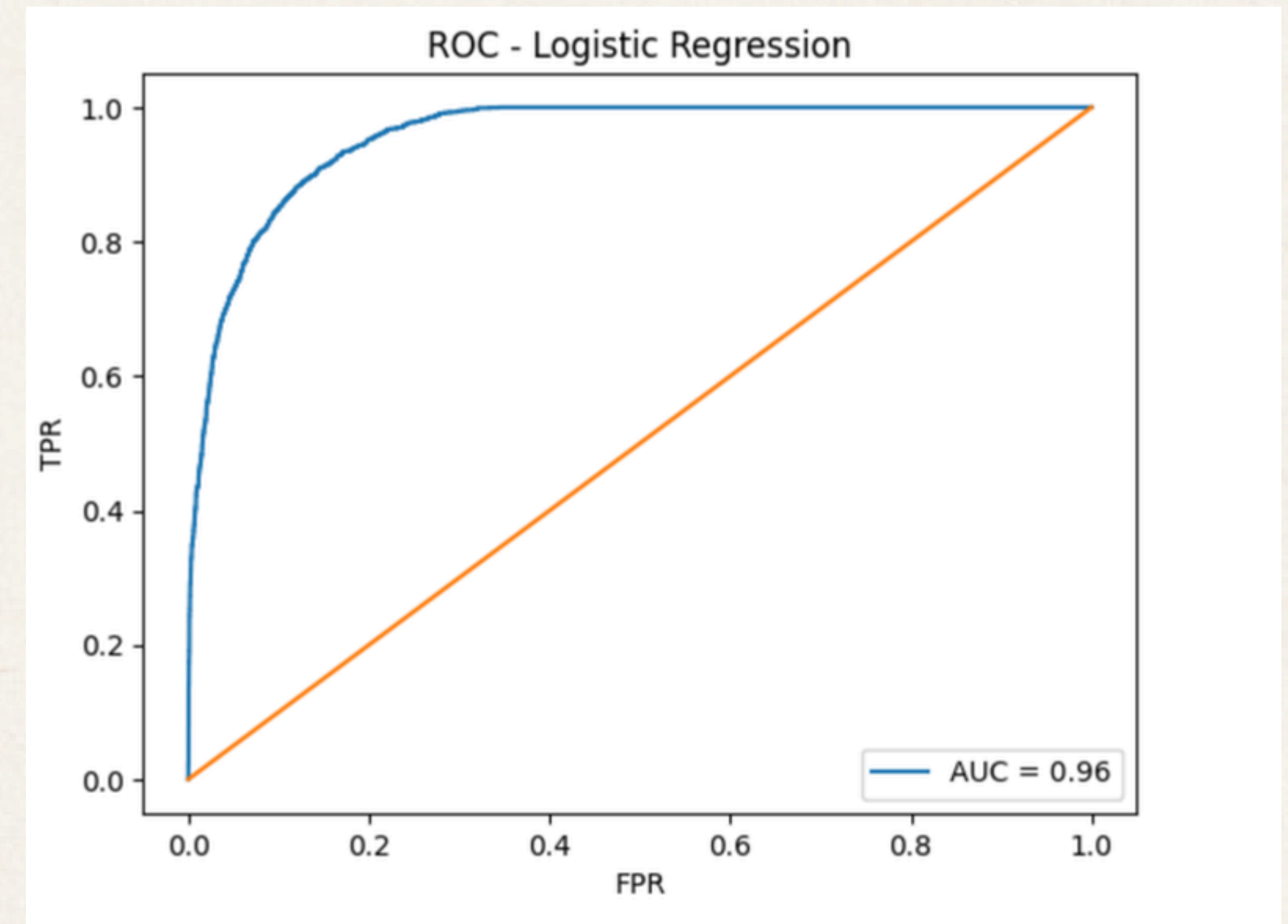
Logistic Regression is a supervised machine learning algorithm used for binary classification problems such as loan approval prediction.

Sigmoid Function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Logistic Model:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$



	precision	recall	f1-score	support
0	0.93	0.94	0.94	7000
1	0.79	0.75	0.77	2000
accuracy			0.90	9000
macro avg	0.86	0.85	0.85	9000
weighted avg	0.90	0.90	0.90	9000

RANDOM FOREST

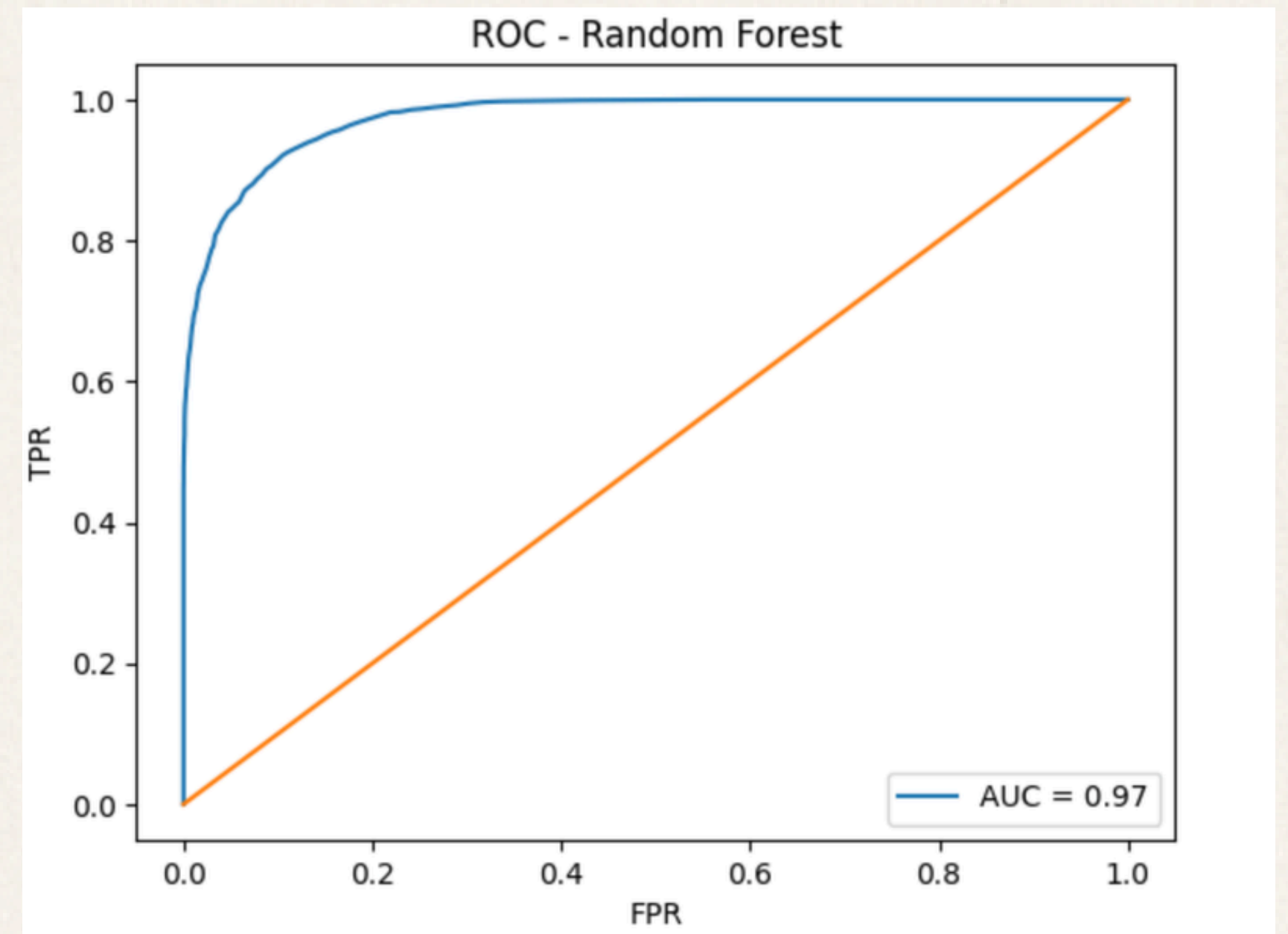
Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. It reduces overfitting by averaging the results of different trees.

Gini Impurity:

$$\text{Gini} = 1 - \sum p_i^2$$

Entropy (Information Gain):

$$\text{Entropy} = -\sum p_i \log_2(p_i)$$



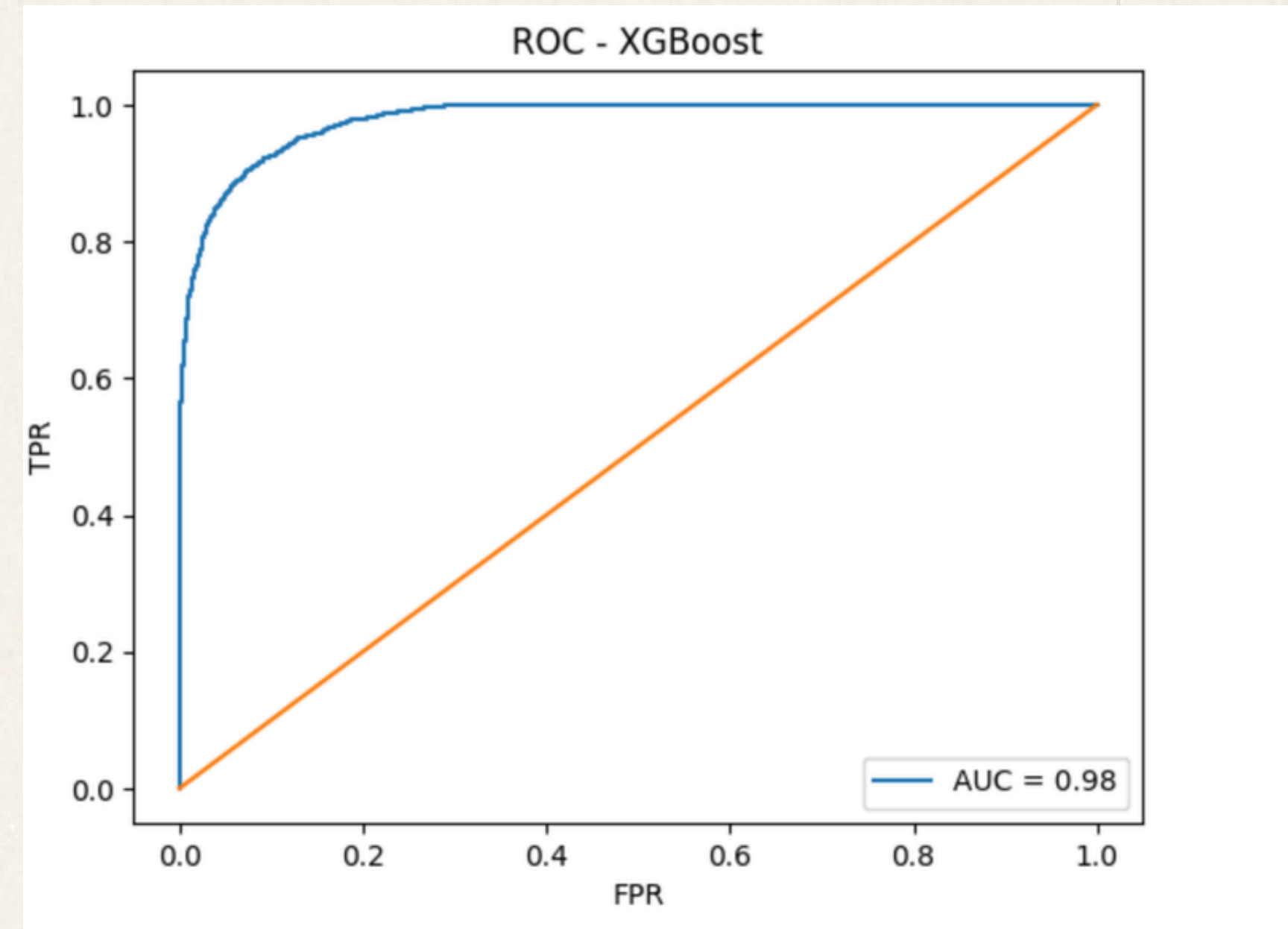
	precision	recall	f1-score	support
0	0.94	0.97	0.95	7000
1	0.90	0.77	0.83	2000
accuracy			0.93	9000
macro avg	0.92	0.87	0.89	9000
weighted avg	0.93	0.93	0.93	9000

XG BOOST

XGBoost is an advanced ensemble learning algorithm based on gradient boosting. It builds decision trees sequentially, where each new tree corrects the errors of the previous one. It provides high accuracy and strong performance on structured/tabular datasets.

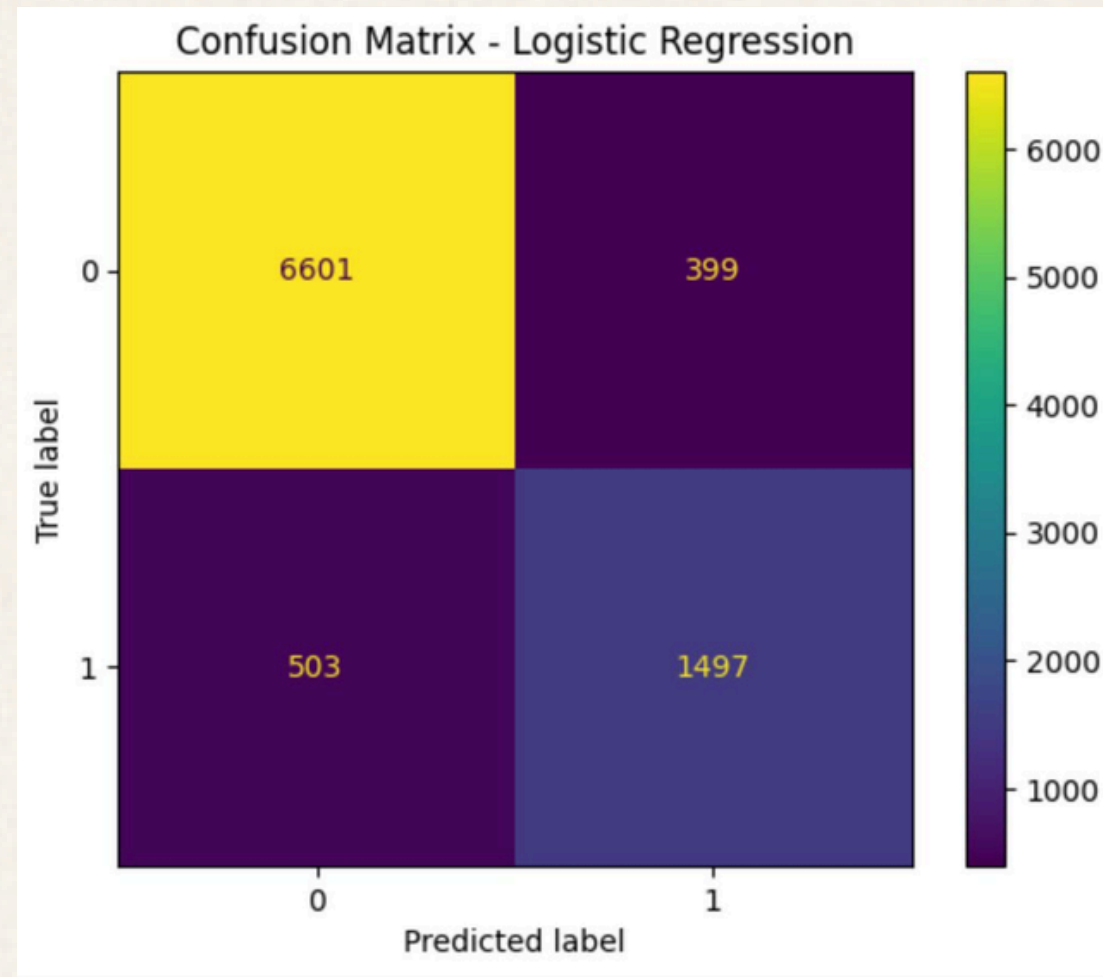
Objective Function:

$$\text{Obj} = \sum L(y_i, \hat{y}_i) + \sum \Omega(f_k)$$

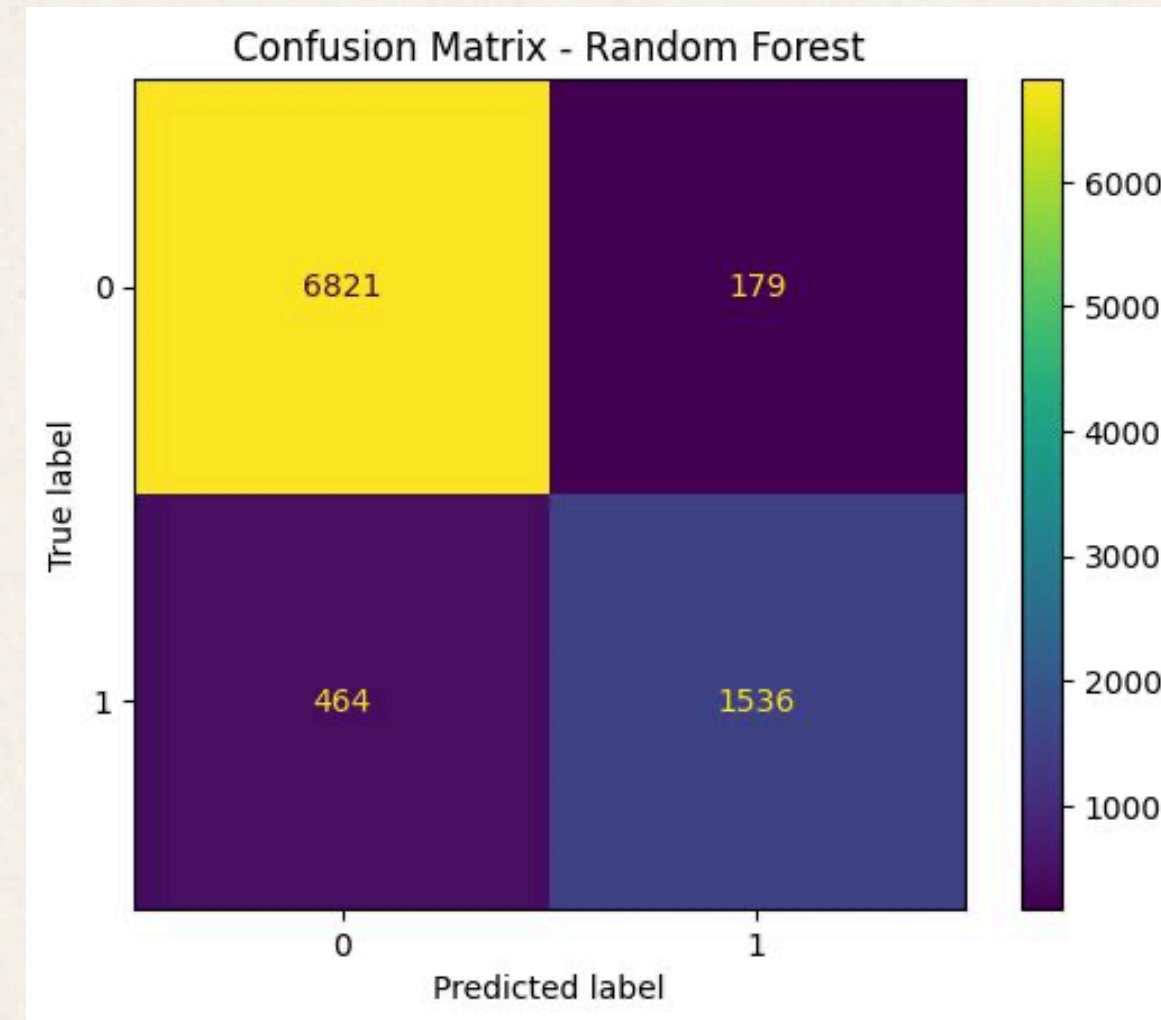


	precision	recall	f1-score	support
0	0.95	0.97	0.96	7000
1	0.89	0.81	0.85	2000
accuracy			0.94	9000
macro avg	0.92	0.89	0.91	9000
weighted avg	0.94	0.94	0.94	9000

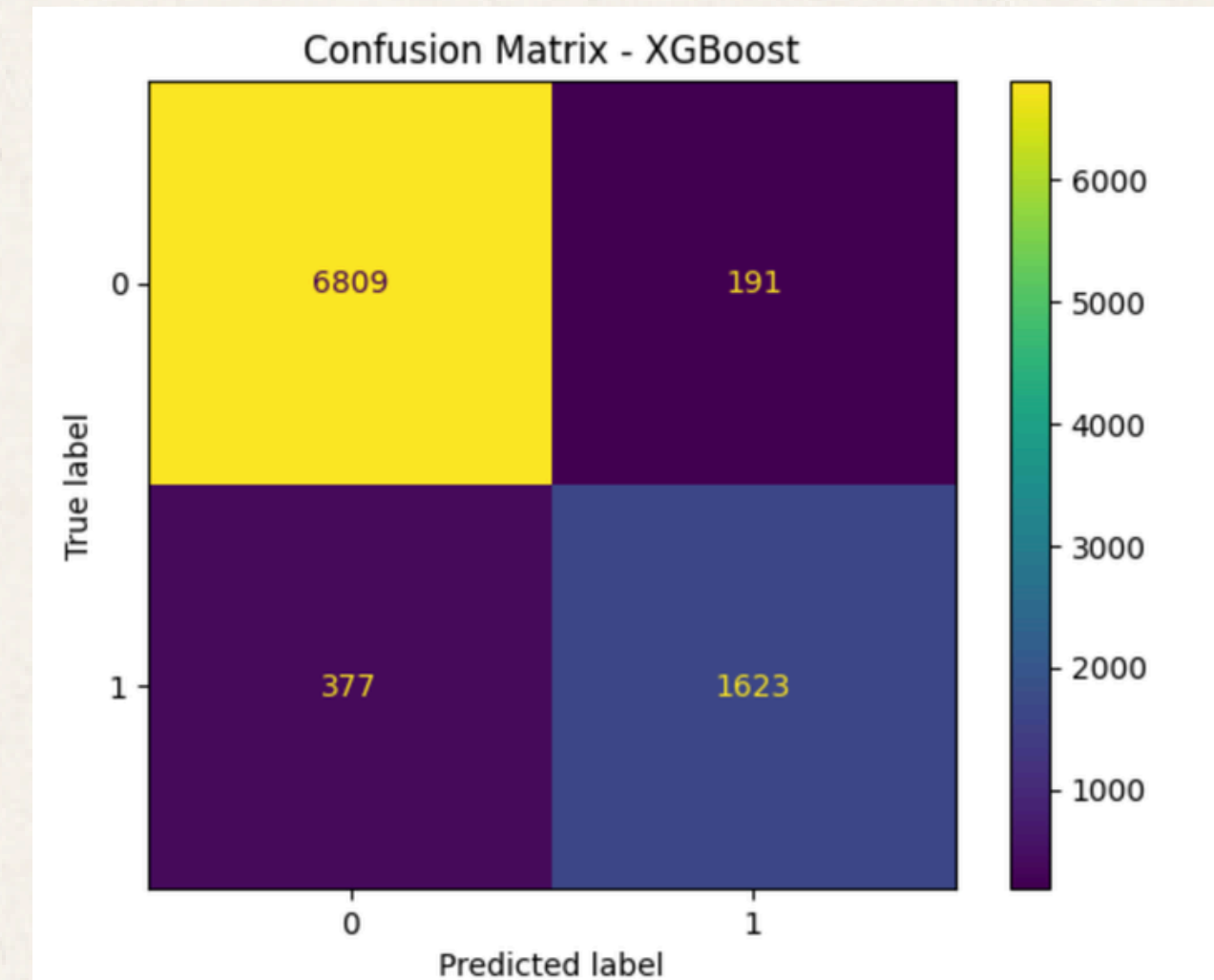
Comparison of models



- **Higher False Negatives (503) and False Positives (399) compared to other models.**
- **Lower True Positives (6601), meaning weaker detection of positive cases.**
- **Overall least accurate among the three models.**



- **Improved True Positives (6821) and reduced False Positives (179).**
- **Fewer False Negatives (464) than Logistic Regression.**
- **Strong performance with better ,balance between precision and recall.**



- **Highest TP+TN (8432) and lowest False Negatives (377).**
- **Very low False Positives (191), close to Random Forest.**
- **Best overall classification performance among all models.**

REFERENCES AND LINKS

Datasets

- <https://www.kaggle.com/datasets>
- <https://www.openml.org>
- <https://data.gov.in>

Models

- <https://www.ibm.com/topics/logistic-regression>
- <https://www.ibm.com/topics/random-forest>
- <https://arxiv.org/abs/1603.02754>